Humanitarian logistics in disaster relief operations

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Abstract

Purpose – This paper aims to further the understanding of planning and carrying out logistics operations in disaster relief.

Design/methodology/approach – Topical literature review of academic and practitioner journals.

Findings – Creates a framework distinguishing between actors, phases, and logistical processes of disaster relief. Drawing parallels of humanitarian logistics and business logistics, the paper discovers and describes the unique characteristics of humanitarian logistics while recognizing the need of humanitarian logistics to learn from business logistics.

Research limitations/implications – The paper is conceptual in nature; empirical research is needed to support the framework. The framework sets a research agenda for academics.

Practical implications – Useful discussion of the unique characteristics of humanitarian logistics. The framework provides practitioners with a tool for planning and carrying out humanitarian logistics operations.

Originality/value – No overarching framework for humanitarian logistics exists in the logistics literature so far. The field of humanitarian logistics has so far received limited attention by logistics academics.

Keywords Logistics data processing, Disasters, Emergency measures

Paper type Research paper

1. Introduction

As a result of the Asian tsunamis in 2004, humanitarian logistics has received increasing interest both from logistics academics as well as practitioners. Humanitarian logistics is an umbrella term for a mixed array of operations. It covers disaster relief as well as continuous support for developing regions. Unfortunately disaster relief will continue to expand market, as it is forecasted that over the next 50 years, both natural and man-made disasters[1] will increase five-fold (Thomas and Kopczak, 2005). Delivering humanitarian aid can, therefore, be seen as a substantial global industry. According to Long and Wood (1995), food relief alone accounted for $5 billion worth of food in 1991; which has important consequences for the world's agricultural and transportation industries. In addition, Long and Wood (1995) estimated the number of major relief agencies at over 100 in 1995, with each of them managing annual budgets of over $1 million. In 2004, the combined budgets of the top ten aid agencies exceeded 14 billion dollars (Thomas and Kopczak, 2005). Also, almost every government in the world is involved as either a donor or recipient of relief operations (Long and Wood, 1995).

Logistics has always been an important factor in humanitarian aid operations, to the extent that logistics efforts account for 80 percent of disaster relief (Trunick, 2005b). The speed of humanitarian aid after a disaster depends "on the ability of logisticians to procure, transport and receive supplies at the site of a humanitarian relief effort"

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(Thomas, 2003, p. 4). But disaster relief operations struggle with very special circumstances. They often have to be carried out in an environment with destabilized infrastructures (Cassidy, 2003; Long and Wood, 1995) ranging from a lack of electricity supplies to limited transport infrastructure. Furthermore, since most natural disasters are unpredictable, the demand for goods in these disasters is also unpredictable (Cassidy, 2003; Murray, 2005). Therefore, a set of characteristics can be identified that sets humanitarian logistics apart from business logistics.

The focus of this paper is on discovering and describing the unique characteristics of humanitarian logistics in disaster relief operations. The paper aims to further the understanding of planning and carrying out logistics operations in the complex area of disaster relief. A framework is created that distinguishes the actors, phases, and logistical processes of disaster relief. The paper commences with a discussion on how the literature review, which lays the foundation for the description of humanitarian logistics, was conducted. Next, the paper discusses the different actors involved in the supply network of humanitarian aid. It concludes with a framework for humanitarian logistics in disaster relief. The concluding discussion also recommends the types of further research needed in this field.

2. Methods

While humanitarian aid efforts date back to the development of social structures and the caring nature of humans, the academic literature on humanitarian logistics is scant. To date there are no dedicated journals to humanitarian logistics, and there is only a limited body of research on the topic (Beamon and Kotleba, 2006). In addition, most of this literature is not academic but originates in practitioner journals. Even though an article on famine relief was found in the *Journal of Business Logistics* in 1995 (Long and Wood, 1995), the overwhelming number of articles on humanitarian logistics were published in practitioner journals. This indicates a need for more academic research in the field. Nonetheless, articles in practitioner journals also give an insight into an emerging field and are helpful in gaining and initial understanding of this field. Therefore, it was deemed necessary to conduct a literature review prior to developing a framework for disaster relief logistics.

This literature review took the form of a series of keyword searches in several journal databases. Keyword searches were chosen as a method for sampling literature instead of a content analysis, because there are no journals dedicated to humanitarian logistics, and academic research in the field is quite limited. Relevant keywords were derived from prior articles and from suggestions provided in the databases. The following keywords were used for the literature review:

- “humanitarian” and “logistics;”
- “humanitarian aid” and “supply chains;”
- “disaster relief” and “logistics;”
- “disaster relief” and “supply chains;”
- “disaster recovery” and “supply chains;”
- “emergency “ and “logistics;” and
- “emergency” and “supply chains”[2].

The literature review was then gradually expanded by using the reference lists of the articles found using the keyword search. In addition to conducting the search of journals, a general internet search was conducted with the same search terms used in the journal search. This search provided an interesting broadening of the area, as we found articles and information from several governmental agencies and relief organizations as well as other institutes and organizations that have taken an interest in the field for a long time.

3. Humanitarian logistics

Humanitarian logistics encompasses very different operations at different times, and as a response to various catastrophes. All these operations have the common aim to aid people in their survival. Nonetheless, aid to assist the development of a region, famine aid and the running of refugee camps is substantially different from the kind of aid needed after a natural disaster. Thus, two main streams of humanitarian logistics can be distinguished, continuous aid work, and disaster relief. While famine relief is sometimes also covered under "disaster relief" (Long, 1997), usually, the term disaster relief is reserved for sudden catastrophes such as natural disasters (earthquakes, avalanches, hurricanes, floods, fires, volcano eruptions, etc.) and very few man-made disasters such as terrorist acts or nuclear accidents. Relief itself can be defined as a "foreign intervention into a society with the intention of helping local citizens" (Long and Wood, 1995, p. 213). The focus of disaster relief operations is to:

> . . . design the transportation of first aid material, food, equipment, and rescue personnel from supply points to a large number of destination nodes geographically scattered over the disaster region and the evacuation and transfer of people affected by the disaster to the health care centers safely and very rapidly (Barbarosoğlu *et al.*, 2002, p. 118).

Disaster management is often described as a process with several stages (Long, 1997; Nisha de Silva, 2001). Cottrill (2002), borrowing from the risk management literature, talks about the planning, mitigation, detection, response and recovery phases of disaster management. Adopting this to the needs for information technology in humanitarian logistics, Lee and Zbinden (2003) discuss three phases of disaster relief operations, the phases of preparedness, during operations, and post-operations. Thus, different operations can be distinguished in the times before a disaster strikes (the preparation phase), instantly after a disaster (the immediate response phase) and in the aftermath of a natural disaster (the reconstruction phase) (Figure 1). In Long's (1997) terms, the first two phases correspond to strategic planning to prepare for emergency projects, and actual project planning when disaster strikes.

Not surprisingly, different resources and skills are needed for the three distinct phases of disaster relief. Therefore, the following discussion will further expand the phases of disaster relief as shown in Figure 1.

3.1 Preparing for a disaster

While natural disasters are difficult to prevent, some regions are more prone to them than others and can thus prepare for particular risks. Tokyo, San Francisco and

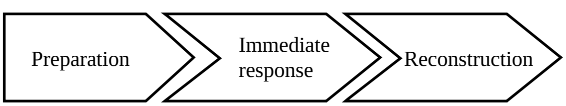

Figure 1.
Phases of disaster relief operations

Reykjavík are examples of cities that need to carefully prepare for the possibilities of major earthquakes; other cities and regions are too close to an active volcano to ignore evacuation plans, or lie in hurricane-prone regions. Avalanches, on the other hand, are natural disasters that can even be prevented by offering, e.g. free training. Evacuation plans can be developed and evacuation can be trained well in advance for such disasters (Nisha de Silva, 2001). Also, measures can be taken to limit the effects of disasters. As an example, providers of energy in hurricane-prone areas can put their connections under ground, thus minimizing the risk of power shortages and even the number of electrocuted victims (Longo, 2005). However, as donors insist that their money goes directly to help victims and not to finance back-office operations, preparation and training are often neglected (Murray, 2005). For obvious reasons, regional actors, e.g. regional governments, businesses and non-governmental organizations, should prepare such plans. According to Chaikin (2003), emergency preparedness plans unfortunately often lack any insight in disaster relief logistics.

What was found, however, is that the academic literature on the topic tackles many different problems of disaster relief, and that this literature actually focuses on the preparation phase. Several decision support systems and technologies have been developed for disaster scenarios at particular sites. These include spatial decision-support systems (Nisha de Silva, 2001), the creation of realistic disaster scenarios and their validation (Nisha de Silva, 2001), simulation techniques, vehicle routing problems for emergency situations (Barbarosoğlu *et al.*, 2002; Özdamar *et al.*, 2004) and distribution problems (Hwang, 1999). What these problems have in common is that they assume particular scenarios and the existence of input data such as known nodes of demand for particular goods. Few of them concentrate on the second phase, the immediate response after a natural disaster, and even fewer deal with the dynamic situation of emergencies (Özdamar *et al.*, 2004). Instead they focus on evacuation plans for the case when a disaster can be predicted (Nisha de Silva, 2001) or focus on the mathematical solutions for vehicle routing models (Barbarosoğlu *et al.*, 2002; Hwang, 1999). Logistical support is needed in prevention and evacuation-related measures before a disaster strikes (which might be foreseen, e.g. when volcanoes are predicted to erupt or hurricanes are approaching a region); in instant medical and food relief procedures once a disaster strikes, and during reconstruction phases.

Apart from measures to prevent disasters, strategic plans can also be developed for the case when they do occur (Long, 1997). According to Thomas (2003), logistics actually serves as a bridge between disaster preparedness and response. Some items are so frequently needed in natural disasters that aid agencies typically develop strong relationships with their suppliers and have long-term purchasing agreements. Thus, UNICEF's disaster management distribution centre collects most commonly needed items continuously in Copenhagen (Dignan, 2005). In essence, stock can be pre-positioned (Thomas, 2003). According to Dignan (2005) goods that are most commonly needed in disaster relief are water, medicine, chlorination tablets, tents, blankets and protein biscuits for malnourished children. Many relief agencies indeed have pre-purchasing agreements with suppliers of drugs, tents, sheeting or blankets (Murray, 2005). Information technology is also crucial to humanitarian efforts. Long (1997) argues that information systems are the single most important factor in determining the success or failure of a disaster relief operation. Regional actors need accurate information to create realistic disaster scenarios upon which they can develop, e.g. evacuation plans (Nisha de Silva, 2001). Also, hospitals in disaster-prone regions who track their particular needs at the times of disasters can develop emergency purchasing procedures with their suppliers (DeJohn, 2005).

The coordination of many different aid agencies, suppliers, and local and regional actors, all with their own ways of operating and own structures can be very challenging (Long and Wood, 1995). A lack of coordination often leads to confusion at the last mile (Murray, 2005). Therefore, the preparation phase is also the time in which aid agencies can develop collaborative platforms such as the United Nations Joint Logistics Centre (UNJLC) (Kaatrud *et al.*, 2003) or the Disaster Relief Network operated by the World Economic Forum (Bradley *et al.*, 2002; Sawyer *et al.*, 2005). At the same time, coordination software is being developed for the special purposes of humanitarian logistics (Tomasini and van Wassenhove, 2004), e.g. the Humanitarian Logistics Software of the Fritz Institute (Murray, 2005). Recently, many "traditional" transportation companies such as DHL or TNT logistics have entered the scene of disaster relief operations, establishing partnerships with the UN.

3.2 Immediate disaster response

Once a disaster strikes, the emergency plans of regional actors come to action. But, however, prepared these actors are, they will need to operate in an environment with a destabilized infrastructure (Cassidy, 2003; Murray, 2005). Moreover, some disasters such as famines occur more often in less developed regions, which from the outset struggle with inadequate infrastructures and a lack of transport connectivity (Long and Wood, 1995). Less developed regions are also more prone to a larger scale destruction of their infrastructure once a disaster strikes. As an example, earthquakes and floods are often magnified, due to poor housing situations and inadequate construction requirements.

The nature of most disasters demands an immediate response, hence supply chains need to be designed and deployed at once even though the knowledge of the situation is very limited (Beamon, 2004; Long and Wood, 1995; Tomasini and van Wassenhove, 2004). Business logistics usually deals with a predetermined set of suppliers, manufacturing sites, and a stable or at least predictable demand – all of which are unknown in humanitarian logistics (Cassidy, 2003). Descriptions of disaster relief operations frequently criticize aid agencies for their lack of collaboration and coordination (McClintock, 2005; Murray, 2005; Sowinski, 2003). There is an abundance of aid agencies focusing on relief after natural disasters (Long and Wood, 1995). Therefore, it is often unknown which resources are available, and even the involvement and contribution of suppliers is unpredictable (Tomasini and van Wassenhove, 2004). This creates many redundancies and duplicated efforts and materials (Simpson, 2005). While military relief operations are usually coordinated from one particular coordination centre (Özdamar *et al.*, 2004; Roosevelt, 2005), the involvement of many different aid agencies in relief operations renders many distribution centre-based planning techniques obsolete. Long (1997) goes so far as to argue against the use of centralized distribution facilities, as victims are often weakened and cannot travel long distances to receive aid. Given all the challenges to coordinate a multi-facility and multi-supplier network, a major emphasis is placed on real-time communication in disaster relief operations (Long and Wood, 1995).

In the immediate response phase, remote aid agencies assume the needs of disaster victims based on very limited information (Long and Wood, 1995). Assumptions need to be made regarding the kind and quality of supplies needed, the times and locations of demand, as well as the nature of the potential distribution of these supplies to any point of demand (Long and Wood, 1995). In fact, the main problem areas of the immediate response phase lie in coordinating supply, the unpredictability of demand, and the last mile problem of transporting necessary items to disaster victims (Beamon, 2004; Long, 1997; Long and Wood, 1995; Özdamar *et al.*, 2004; Tomasini and van Wassenhove, 2004). According to Ernst (2003), three major processes can be distinguished also when structuring and analyzing commercial logistics: demand management, supply management, and fulfillment management. In the following, we will relate these phases to humanitarian logistics.

3.2.1 Demand management. Assessing demand after a disaster also includes a consideration of the cultural peculiarities of the disaster region (Beamon, 2004; Trunick, 2005c, Wichmann, 1999). Language barriers in a disaster region also complicate the distribution of adequate supplies (Long and Wood, 1995). Demand is unpredictable regarding timing, location, and scale (Beamon, 2004; Murray, 2005; Long, 1997; Long and Wood, 1995). As Arminas (2005, p. 14) puts it:

> . . . purchasing and logistics for major disaster relief is like having the client from hell – you never know beforehand what they want, when they want it, how much they want and even where they want it sent.

A particular problem occurs when traditional relief destinations such as hospitals are destroyed through the disaster (DeJohn, 2005). Hoffman (2005) notes that humanitarian supply chains are the most dynamic supply chains in the world. In emergencies, vehicles are called from every node in a network, independent of the actual demand in the node (Özdamar *et al.*, 2004). This can lead to trucks circulating around a disaster area without any particular destination (Greiling Keane, 2005):

> Given the unpredictability of demand and the limited information aid agencies have in the first hours and days after a disaster, supplies are "pushed" to the disaster location in a first phase (Long and Wood, 1995). Only in later stages of the relief operation can more accurate data on the needs of disaster victims be assembled, changing push supplies to more pull operations (Long and Wood, 1995).

3.2.2 Supply management. In terms of supply, aid agencies receive many unsolicited and sometimes even unwanted donations (Chomolier *et al.*, 2003). These can include drugs and foods that are past their expiry dates (Murray, 2005); laptops needing electricity which infrastructure has been destroyed; heavy clothing not suitable for tropical regions (Dignan, 2005), etc.:

> Inappropriate donations are so common that relief missions now routinely bring incinerators with them to the scene of a disaster to destroy items that may be dangerous or are clogging up the system (Murray, 2005, p. 9).

Unsolicited supplies in fact clog airports and warehouses (Cassidy, 2003; Murray, 2005) and create redundancies (Sowinski, 2003).

Even solicited supplies arrive in unmanageable forms. Aid agencies involved in a disaster relief operation originate from many different countries, and donors send their items with labels in a variety of forms and languages. The lack of standard labeling of supplies is indeed one of the biggest problems of distributing aid at sites (Murray, 2005). Therefore, relief agencies started to color-code items, such as using red for foodstuffs and blue for clothing (Murray, 2005). In coordinating supply, aid agencies can fall back on local suppliers (Murray, 2005). In fact, retailers have often been the first aid workers reaching a disaster struck location (Garry, 2005a, b; Leonard, 2005; Rowell, 2005). They have the advantage of reduced transportation needs and are very likely to fulfill the dietary requirements of the regional population.

Recently, UN agencies agreed to join forces in developing a logistics support system (LSS) that will improve coordination at national or international levels among all interested humanitarian partners as well as develop a local capacity. The LSS is a joint instrument available to all institutions that will minimize duplication and improve the response to the actual needs of the affected population while building on the management capacity of institutions. The LSS is built on the experience of a large number of institutions and aims to facilitate the exchange of information among humanitarian agencies. It will complement agency-specific commodity tracking systems that are increasingly developed by larger humanitarian actors (UNJLC, 2005).

3.2.3 Fulfillment management. Kaatrud *et al.* (2003) propose a humanitarian logistician's checklist when discussing the UNJLC. In this checklist, 11 out of 14 points are comprised of infrastructure-related questions – the existence of airports and roads, the availability of vehicles and fuel. Fuel shortages in fact develop very quickly in disaster areas (Sullivan, 2005), sometimes just because fuel-pumping stations are demobilized during power shortages (Anonymous, 2005). Transportation itself is not the biggest problem in disaster relief operations, as airdrops of supplies is always a last option to deliver the necessary goods to disaster victims (Wichmann, 1999). However, often there is a shortage of materials handling equipment at the receiving end (Trunick, 2005a). Packages thus need to be small so they can be handled by a single person (Long and Wood, 1995; Murray, 2005).

The last mile problem, however, poses other issues. A special problem in supplying food in, e.g. famine areas is to insure food safety and hygiene (Gaboury, 2005). Also, much of the medication needed in disaster areas is in need of temperature control – which, given the lack of infrastructure and power supplies, is always a challenge.

3.3 Reconstruction

After the immediate responses, regional actors can begin to aid victims in the location of their family and friends (Lamont, 2005). Unfortunately for many disaster-struck areas, funding is often focused on short-term disaster relief (Gustavsson, 2003). Thus, the long-term phase of reconstruction is neglected. On the other hand, aid agencies, such as World Vision whose mandate is to respond in some way to any disaster around the world, has created a phased relief response which typically occurs in three phases: seven-day, 30-day, and 90-day. During the first phase of the emergency, e.g. flyaway kits are provided. These can sustain up to 2,000 people for seven days. The second phase involves sending family survival kits, which can support up to 5,000 people for 30 days. The third phase is related to reconstruction and it involves long-term rehabilitation. For example, in the aftermath of the earthquake in El Salvador, reconstruction assistance was provided by fixing damaged homes and also by constructing new homes for displaced families.

As pointed out, the reconstruction phase is important as disasters can have long-term effects on a region. In addition, disasters can also have long-term effects on the management of companies. For example, after Hurricane Katrina transportation companies experienced a modal shift from road to rail that still prevails today (Levans, 2005), and some ports are still not operational or have suffered from a reduction in throughput volumes. Therefore, in general it can be argued that regional actors should also focus on the reconstruction phase for which continuity planning is needed. Their disaster prevention plans need to be revised to include things that have been learned from the current disaster (Thomas, 2003).

4. The supply network of humanitarian aid

The examination of the different perspectives on disaster relief operations provides a basis for examining what differentiates humanitarian logistics from business logistics. The first issue concerns the definition of the concept. If we define commercial (business) logistics as the process of managing the flow of goods, information and finances from the source to the final customers, humanitarian logistics can be quite easily defined similarly as humanitarian logistics also requires a process for managing the flow of goods, information and finances from donors to affected persons (Ernst, 2003). Humanitarian logistics, as well as business logistics, encompasses a range of activities including preparedness, planning, procurement, transport, warehousing, tracking and tracing and customs clearance (Thomas and Kopczak, 2005). It can, therefore, be concluded that the basic principles of managing the flows of goods, information and finances also remain valid for humanitarian logistics.

4.1 Actors in supply networks of humanitarian aid

The first and fundamental difference is in the motivation for improving the logistics process, i.e. in the case of humanitarian logistics it is required to go beyond profitability (Ernst, 2003). When structuring and analyzing commercial logistics, three main processes are included; demand management, supply management, and fulfillment management (Ernst, 2003). In humanitarian logistics there are, however, many actors that are not linked to the benefits of satisfying demand (Figure 2).

Suppliers have different motivations for participating and customers are not generating a voluntary demand and will hopefully not create a "repeat purchase." The actor network involved in the humanitarian supply chain process also distinguishes the two fields from each other. An important difference is in the fact that the customer actually has no choice, and therefore, "true demand" is not created.

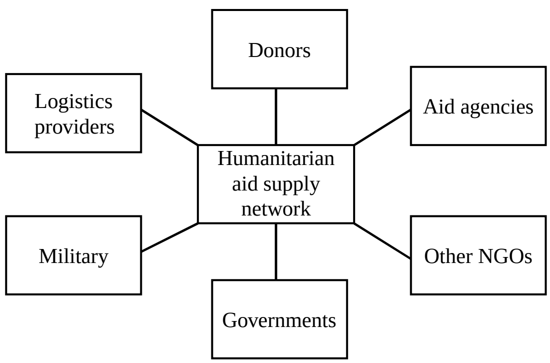

Figure 2.
Actors in the supply network of humanitarian aid

Demand is rather assessed through aid agencies (Long and Wood, 1995), which can be viewed as the primary actors through which governments channel aid that is targeted at alleviating suffering caused by natural and manmade disasters. The largest agencies are global actors, but there are also many small regional and country-specific aid agencies (Thomas and Kopczak, 2005). Many organizations have their own political motives for providing relief (Long and Wood, 1995). Political issues might aggravate the situation of relief operations, and even hinder supplies in reaching a particular region. In some crisis areas such as war zones, rebel forces might even want to block the arrival of these supplies (Murray, 2005). Looting might also occur after natural disasters, and trucks are often stopped and deviated from their intended destination (Cassidy, 2003). The lack of security in these types of operations is striking. Therefore, it is a difficult but important topic to depoliticize relief operations (Tomasini and van Wassenhove, 2004).

Donors are important actors, as they provide the bulk of funding for major relief activities. In addition, to country specific funding (e.g. the USA and EU), in recent years, foundations, individual donors and the private sector have become important sources of funds for aid agencies. Other actors include the military, host governments and neighboring country governments, other non-governmental organizations (NGOs) and logistics service providers (Kaatrud *et al.*, 2003). The military has been in many occasions a very important actor as military personnel are called into provide assistance (Özdamar *et al.*, 2004). For example, the military brought communications, logistics and planning capabilities that were critical to Katrina relief operations. Host governments are important actors as they control assets such as warehouses or fuel depots. Host country logistics or regional service providers are another important set of actors that can either facilitate or constrain the operational effectiveness of humanitarian logistics operations. Extra-regional logistics service providers are also important in the supply process, e.g. DHL has contributed to the international relief efforts to deliver aid supplies to people and communities affected by the South Asia earthquake.

4.2 Characteristics of humanitarian logistics

A set of characteristics that set business logistics apart from humanitarian logistics can now be identified. Business logistics usually deals with a predetermined set of suppliers, manufacturing sites, and stable or at least predictable demand – all of which factors are unknown in humanitarian logistics (Cassidy, 2003). Humanitarian logistics again is characterized by large-scale activities, irregular demand and unusual constraints in large-scale emergencies (Beamon and Kotleba, 2006). In terms of the end-result strived for, business logistics aims at increasing profits whereas humanitarian logistics aims at alleviating the suffering of vulnerable people (Thomas and Kopczak, 2005). The supply network structure of humanitarian logistics also differs from that of business logistics due to the fact that it is comprised of so many actors with no clear or stated linkages to each other. While operations and actors are intertwined, different groups of actors and different phases of disaster relief operations can be distinguished. All these operations have the common aim to aid people in their survival. They often have to be carried out in an environment with destabilized infrastructure (Cassidy, 2003; Long and Wood, 1995) ranging from a lack of electricity to limited transport infrastructure. Furthermore, most natural disasters are unpredictable, thus as a result the demand for goods in these disasters is also unpredictable (Cassidy, 2003; Murray, 2005). The immediate response stage usually involves a large amount of supplies being pushed to the disaster location. A summary of the characteristics of humanitarian logistics that distinguish business logistics from humanitarian logistics is presented in Table I.

Table I highlights the characteristics of humanitarian logistics, however, most of the characteristics can actually be associated with different types of emergency situations, not only those that deal with disaster relief. Nevertheless, what sets this type of an emergency situation apart from others is usually the magnitude of the catastrophe taking place and the logistics operations needed.

4.3 A framework for disaster relief logistics

Generally, the actors involved in disaster relief can be grouped into two large categories, those that exist in the region and are intrinsically linked to it, such as host governments, military, local enterprises and regional aid agencies, and international actors such as the UN, larger aid agencies, extra-regional NGOs, logistics service providers, etc. These different actors take different perspectives on humanitarian logistics, and can in fact prepare and execute disaster relief operations differently. Taking an internal, regional perspective compares disaster prevention and forecasting, even the development of evacuation plans, to risk management (Nisha de Silva, 2001). On the other hand, taking the perspective of aid agencies on preparing for risks in remote regions is seen as strategic project planning (Long, 1997). The different phases of disaster management have been discussed in terms of risk management (Long and Wood, 1995; Zolkos, 2003), crisis management (Nisha de Silva, 2001), strategic and operational planning (Long, 1997), business continuity planning (Mohamed, 2005), and also project management.

Whether disaster management is described in terms of risk and crisis management, or even business continuity planning, or in terms of project planning and execution,

	Humanitarian logistics
The main aim	Alleviating the suffering of vulnerable people
Actor structure	Stakeholder focus with no clear links to each other, dominance of NGOs and governmental actors
3-phase setup	Preparation, immediate response, reconstruction
Basic features	Variability in supplies and suppliers, large-scale activities, irregular demand, and unusual constraints in large-scale emergencies
Supply chain philosophy	Supplies are "pushed" to the disaster location in the immediate response phase. Pull philosophy applied in reconstruction phase
Transportation and infrastructure	Infrastructure destabilized and lack of possibilities to assure quality of food and medical supplies
Time effects	Time delays may result in loss of lives
Bounded knowledge actions	The nature of most disasters demands an immediate response, hence supply chains need to be designed and deployed at once even though the knowledge of the situation is very limited
Supplier structure	Choice limited, sometimes even unwanted suppliers
Control aspects	Lack of control over operations due to emergency situation

Table I.
Characteristics of humanitarian logistics

depends on the perspective taken in a particular disaster relief operation. We, therefore, propose a framework for disaster relief logistics, which separates the perspectives of different actors on a disaster operation during the three different phases of the operation. Figure 3 shows the above analogies in terms of the perspectives taken within and outside the disaster region. The regional perspective refers to the measures taken where the disaster strikes, while the extra-regional perspective comprises the one taken by donors and aid agencies, governments and other actors involved in relief operations.

While there are crucial differences between regional and extra-regional actors in preparing for a disaster, the two groups interact at the moment of immediate disaster response. Logistics is indeed involved in every stage of relief efforts (Thomas, 2003). After Hurricane Katrina, regional retailers played a crucial role in distributing foodstuffs and clothing to disaster victims (Garry, 2005a; Leonard, 2005; Rowell, 2005) and required close cooperation with the military and aid agencies. On a higher level, the states struck by the disaster needed to cooperate with the Federal Emergency Management Agency in planning and executing relief operations. This collaboration is present in all three phases of disaster relief. Aid agencies and federal governments planning for disaster relief need to take regional disaster prevention programs into account. During the immediate response phase, the coordination of the activities of all actors in the supply network of humanitarian aid is of extreme importance. But also in the reconstruction phase, these actors need to coordinate their efforts and collaborate in reviving the infrastructure of a region. Thus, while the different actors involved in the supply network of humanitarian aid take different perspectives on disaster relief operations, their cooperation is of extreme importance for the success of these operations.

The framework in Figure 3 thus combines the perspectives of different actors involved in delivering humanitarian aid on disaster relief operations with the three different phases of the operations. For each actor and phase parallels to business logistics can be drawn. While each actor involved in disaster relief logistics will need to take the contextual differences of humanitarian logistics to business logistics into account, from the difficulties to predict demand to co-ordinate supply, the tools and methods they need in disaster relief can be adapted from business logistics. Risk management will help regional actors in the phase of preparing for disasters,

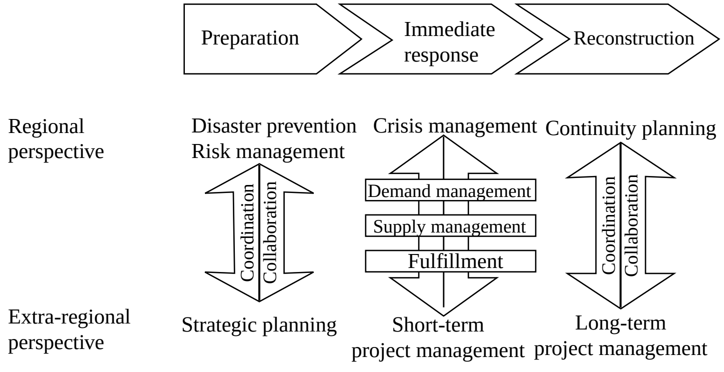

Figure 3.
A framework for disaster relief logistics

while extra-regional actors can turn to strategic planning of disaster relief operations. In the immediate response phase, regional actors can learn from crisis management, or even from the response to disruptions in material flows in business logistics. During this phase, extra-regional actors will employ short-term project management in their part of disaster relief. The last, reconstruction phase of disaster relief logistics is in fact rather similar to a business logistics environment – though not aiming at generating profit. Nonetheless, continuity planning from a regional perspective, and long-term project management from an extra-regional perspective do not have to deal with the same irregularities of demand and supply as the immediate response phase.

5. Concluding discussion

The research field of humanitarian logistics is relatively new. So far no dedicated journal exists, nor have we been able to identify any journal or other outlet focusing on this important topic. Nevertheless, one of the notable aspects of the relief efforts following the Asian tsunami in 2004 was that logistics was publicly acknowledged to play an extremely important role in relief logistics (Thomas and Kopczak, 2005). Yet, many organizations continue to underestimate the importance of logistics in disaster relief operations and still focus on fundraising activities only (Murray, 2005). Academic literature on humanitarian logistics tends to concentrate on the preparation phase of disaster relief. Donors, on the other hand, focus on the immediate response phase after a disaster. Nonetheless, as discussed in our framework, all three phases of humanitarian logistics, preparation, immediate response, and reconstruction, are important in overcoming a disaster. This paper therefore fills a void in reviewing the field both in terms of academic and practitioner contributions, and aids in furthering the understanding of logistical operations in disaster relief.

According to Sowinski (2003), learning from humanitarian logistics will be important for business logistics and commercial supply chains, as disaster relief operations show how to manage unpredictable environments. Based on our literature review, this statement does not hold true, and there is a consensus among field experts that there are many lessons and practices from the commercial world that could be used in the humanitarian world (Ernst, 2003). It is argued that the aid sector is old-fashioned and that it still regards logistics as a necessary expense. It also lacks operational knowledge and has insufficient investments in technology and communication as well as knowledge of the latest methods and techniques, e.g. mathematical modeling (Gustavsson, 2003; Beamon and Kotleba, 2006). There is also a shortage of logistics experts; the supply chain processes are largely manual: there is inadequate assessment and planning and limited collaboration and coordination. Further research is needed in the field of humanitarian logistics, in order to support the planning and execution of the important operations of disaster relief. In this paper, a framework is proposed that illustrates the links between different actors and phases of disaster relief operations. This framework also draws parallels between topics in humanitarian logistics, and business logistics. Although these parallels to risk management, crisis management, continuity planning and project management are already indicated in literature, further research is necessary to examine each of these links and propose their specific implications for humanitarian logistics.

Apparently, the challenges in this field are still vast. Help is, however, on its way as researchers and practitioners, as well as aid organizations and governments, have initiated different forms of cooperation. A good example of this is the Humanitarian and Emergency Logistics Forum (HELP) in the UK, which brings together logistics professionals, organizations and foundations and other groups to leverage expertise, experience and resources in all phases of relief operations.

In conclusion it can be stated that although humanitarian logistics has its distinct features, the basic principles of business logistics can be applied. The principles of the HELP forum nicely exemplifies this as they combine in their aims business logistics principles with the altruistic motivations of humanitarian logistics: "Right people, equipment and material, in the right place, in the right sequence as soon as possible, to deliver the maximum relief at the least cost – saved lives, reduced suffering and the best use of donated funds" (CILT, 2006).

Notes

1. Within disaster relief, a distinction can be made between man-made disasters such as wars and terrorism-related catastrophes, and natural disasters such as floods, fires or earthquakes.
2. The reference list of this paper does not encompass all the articles found in the keyword searches. An extensive reference list can be obtained from the authors on request.

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